

BRODIE E. MANGAN

PhD Candidate, Cognitive Psychology
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RESEARCH INTERESTS

Working Memory, Attention, Cognitive Control, Temporal Cognition, Neural Dynamics, EEG, Time-Frequency Analysis, Cross-Frequency Coupling, Signal Detection Theory, Computational Approaches to Cognition, Rhythm Perception.

EDUCATION

University of Stirling <i>PhD in Psychology</i> Thesis: Cognitive Fatigue and Working Memory: Theory, Measurement, and Methodological Implications for Intervention Research Primary Supervisor: Dr Dimitrios Kourtis Secondary Supervisor: Dr Simone Tomaz	<i>Stirling, UK</i> <i>Sept 2023 – Present (expected completion: Autumn 2026)</i>
University of Glasgow <i>MSc in Brain Sciences (Merit)</i>	<i>Glasgow, UK</i> <i>2021</i>
Glasgow Caledonian University <i>Graduate Diploma in Psychology (Distinction)</i>	<i>Glasgow, UK</i> <i>2020</i>
The Open University <i>Investigating Psychology 1 (DE100) (Distinction)</i>	<i>Distance Learning</i> <i>2019</i>
University of the West of Scotland <i>BA in Sports Development</i>	<i>Hamilton, UK</i> <i>2012</i>

PUBLICATIONS AND RESEARCH OUTPUT

PEER-REVIEWED ARTICLES

MANGAN, B. E., & Kourtis, D. (2025). The Missing Link: Bridging Cognitive Fatigue with Working Memory. *Journal of Cognitive Neuroscience*. <https://doi.org/10.1162/JOCN.a.2398>

MANGAN, B. E., Tomaz, S., & Kourtis, D. (2026). The Missing Link: WAND for Active Cognitive Fatigue Induction. Under review at *PLOS ONE*.

MANGAN, B. E. (2026). Operational Drift: Why Validated Fatigue Induction is the Missing Link in Nature Restoration Research. *Frontiers in Environmental Psychology*. <https://doi.org/10.3389/fpsyg.2026.1780136>

SOFTWARE

MANGAN, B. E. (2026). WAND (Working-memory Adaptive-fatigue with N-back Difficulty): A Modular Software Suite for Cognitive Fatigue Research. Forthcoming at *Journal of Open Source Software*.

PRESENTATIONS AND INVITED TALKS

(2026, June). *Does cognitive fatigue disrupt the theta-gamma code? An EEG test of the missing link hypothesis*. Oral presentation submitted to the European Society for Cognitive and Affective Neuroscience (ESCAN), Rome, Italy.

(2026, April). Poster presentation accepted to the British Neuroscience Association Members Meeting (BNA). Online.

(2025, September). *Validating WAND, a Protocol to Isolate Active Cognitive Fatigue*. Poster presentation at the Annual Meeting of the British Association for Cognitive Neurosciences (BACN), University of Edinburgh, UK.

(2024, May). *Laying the Groundwork: Objective Fatigue Measurement for Assessing Green Space Cognitive Restoration*. Invited talk, Psychology Research Seminar, University of Stirling, UK.

(2024, April). *Cognitive Enhancement via Green Space Environments*. Invited talk, Institute of Advanced Studies (Accessible Environments Cluster), University of Stirling, UK.

HONOURS AND AWARDS

European Society for Cognitive and Affective Neuroscience (ESCAN) 2026 – Travel Grant Award.
British Association for Cognitive Neuroscience Conference 2025 – Poster Prize Winner.
Nominated for ‘Outstanding Early Career Researcher’, University of Stirling Research Culture Awards | 2025.

SKILLS

RESEARCH METHODS. Experimental design, human behavioural data collection, signal detection theory, computational modelling, mixed-effects models, reproducible analysis workflows.

PROGRAMMING AND SOFTWARE. Python, PsychoPy, R/RStudio, custom experimental platforms, reproducible pipelines, open-source research tools.

EEG AND NEUROPHYSIOLOGY. EEG acquisition and preprocessing, time-frequency analysis, event-related potentials, cross-frequency coupling, spectral parameterisation, frequency-domain analysis.

LANGUAGES. English (native). Keen to learn French.

ADDITIONAL RELEVANT EXPERIENCE

Longstanding personal engagement with music, including approximately ten years of hobbyist sample-based music production and broad interest in live music across classical and electronic genres.

PUBLIC ENGAGEMENT

Outreach and Public Engagement Training, Glasgow Science Centre (2025). Developed and delivered hands-on activities about space telescopes and exoplanet discovery to over 200 visitors.

RELEVANT PROFESSIONAL EXPERIENCE

Personal Trainer and High-Performance Coach 2008 – Present
Self-employed, Glasgow

- Developed and delivered long-term training and nutrition programmes for 1,000+ clients, ranging from beginners to elite athletes (part-time from 2023).
- Provided high-performance coaching for Red Bull projects, including multi-year work with Danny MacAskill across project completion, injury recovery, and commercial productions.
- Translated complex evidence into clear action plans and tracked adherence and outcomes over long time horizons.

Personal Health Consultant (Occupational Health) 2014 – 2018
EDF Energy, Hunterston Nuclear Power Station

- Designed and delivered data-driven lifestyle and training interventions to improve staff health and performance metrics in a high-risk industrial setting.
- Contributed to a team initiative that won the Health Working Lives Gold Award.
- Delivered coaching and education sessions and monitored engagement over time.

Paratrooper 2004 – 2008
The Parachute Regiment, 16 Air Assault Brigade, British Army

- Completed Pre-Parachute Selection (P Company) and the Basic Parachute Course.
- Qualified as a team medic, VHF/HF radio operator, and machine gun team leader.
- Served on operations in Afghanistan (Op Herrick, Helmand Province) and on exercises worldwide, including with US Special Operations, where awarded US Paratrooper Wings.

REFERENCES

Dr Dimitrios Kourtis (Primary Supervisor) | Senior Lecturer in Psychology, University of Stirling | dimitrios.kourtis@stir.ac.uk | +44 (0)1786 466979

Dr Simone Tomaz (Secondary Supervisor) | Lecturer in Exercise Physiology, University of Stirling | simone.tomaz@stir.ac.uk